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**GOOGLE PROMPT ENGINEERING COURSE**

# PROMPT COMPARISON ENGINE

**NAZSYA AI TUTOR**

**AI Smart Tutor Using Advanced Prompt Engineering Techniques**

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# A. Prompt Comparison Engine and Optimization

## 1. Prompt Comparison Engine Development

In this phase, the project developed a Prompt Comparison Engine to compare the quality of AI-generated educational responses using different prompt engineering techniques.

The initial prototype tested several prompting strategies, including Standard Prompting, Basic Prompting, Structured Prompting, Few-Shot Prompting, and Optimized Prompting. This comparison was used to analyze differences in response quality, structure, consistency, and readability.

In the final OpenRouter-based implementation, the comparison engine focused on comparing the Basic Prompt Response and the Optimized Prompt Response. This approach allowed the project to clearly evaluate how optimized prompt engineering improved the educational quality of the AI tutor response.

### 1.1 Initial Multiple Prompt Generation Implementation

The following code shows the initial prompt comparison implementation used during the early prototype stage.

```
# Generate Responses

standard_response = model.generate_content(prompt)

basic_response = model.generate_content(basic_prompt)

structured_response = model.generate_content(structured_prompt)

few_shot_response = model.generate_content(few_shot_prompt)

optimized_response = model.generate_content(optimized_prompt)
```


**Figure 1.1** *Multiple prompt generation implementation using several prompt engineering techniques*

### 1.2 Prompt Comparison Display Implementation

```
st.subheader("Response 1 — Standard AI Tutor")
st.write(standard_response.text)

st.subheader("Response 2 — Basic Prompt")
st.write(basic_response.text)

st.subheader("Response 3 — Structured Prompt")
```

```
st.write(structured_response.text)

st.subheader("Response 4 — Few-Shot Prompt")
st.write(few_shot_response.text)

st.subheader("Response 5 — Optimized Prompt")
st.write(optimized_response.text)
```

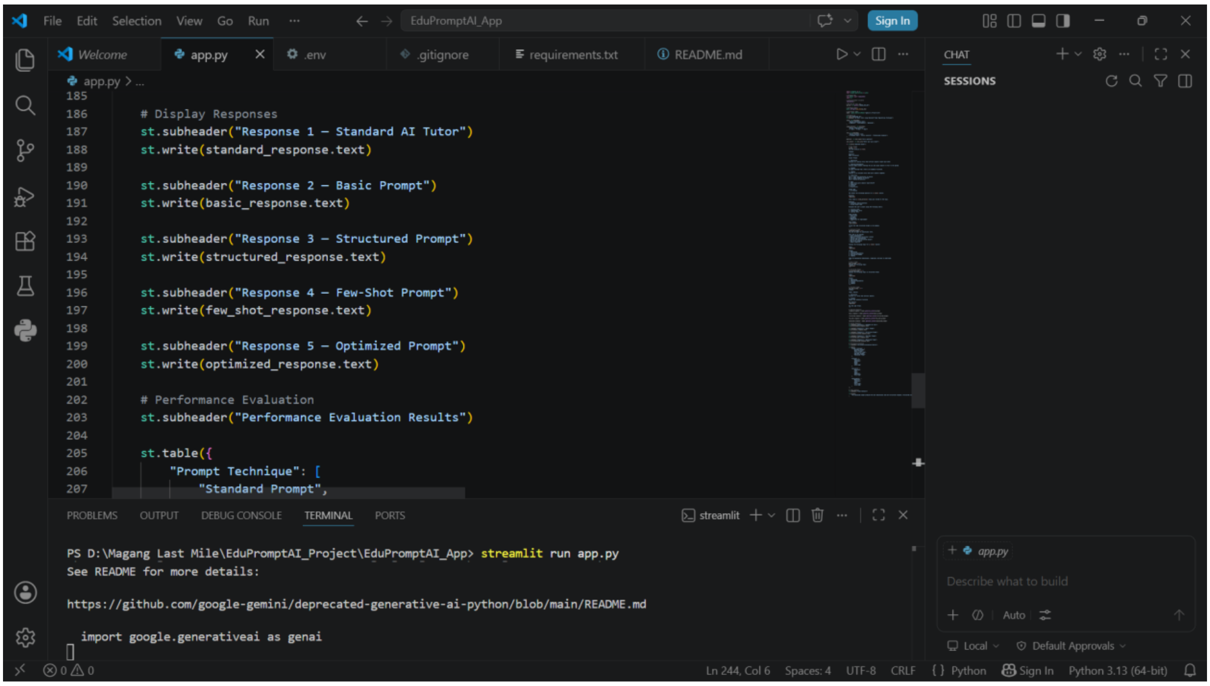



**Figure 1.2** *Comparison of multiple AI-generated responses using different prompt engineering techniques*

## 1.3 Final Prompt Comparison Engine Implementation in OpenRouter Version

After the initial prompt comparison engine was tested, the feature was adapted into the final version of Nazsya AI Tutor using OpenRouter API and the DeepSeek Chat model.

In the final implementation, the system compares two main response types: Basic Prompt Response and Optimized Prompt Response. The Basic Prompt Response generates a short and general explanation, while the Optimized Prompt Response applies multiple prompt engineering techniques, including structured prompting, learner-level adaptation, few-shot prompting, study plan generation, quiz generation, and role-based instructional guidance.

This final comparison helps demonstrate the improvement produced by prompt optimization in terms of structure, completeness, personalization, and educational usefulness.

```python
basic_prompt = f"""
Explain this topic briefly:

{question}
"""

basic_response = client.chat.completions.create(
    model="deepseek/deepseek-chat",
    messages=[
        {
            "role": "user",
            "content": basic_prompt
        }
    ]
)

basic_output = basic_response.choices[0].message.content
```

```python
response = client.chat.completions.create(
    model="deepseek/deepseek-chat",
    messages=[
        {
            "role": "user",
            "content": optimized_prompt
        }
    ]
)

optimized_response = response.choices[0].message.content
```

**Figure 1.3** *Basic prompt response implementation using OpenRouter API*

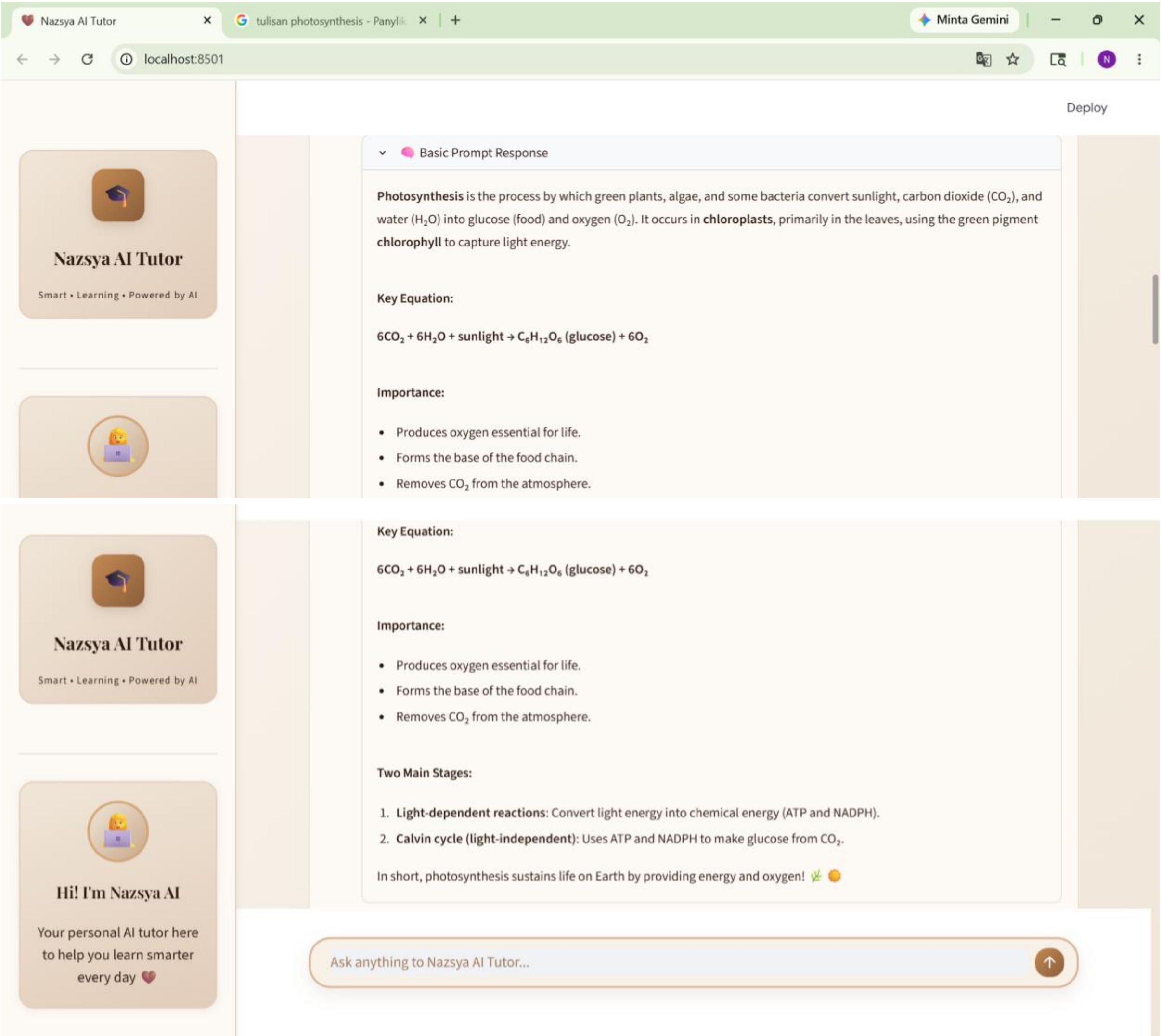
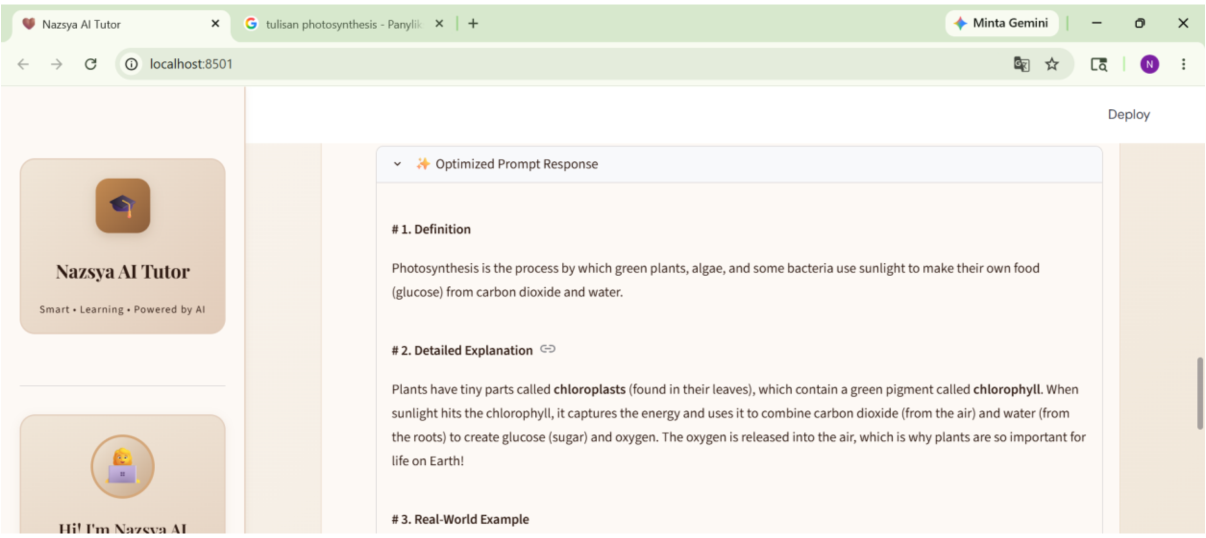



**Figure 1.4** *Optimized prompt response implementation using OpenRouter API*



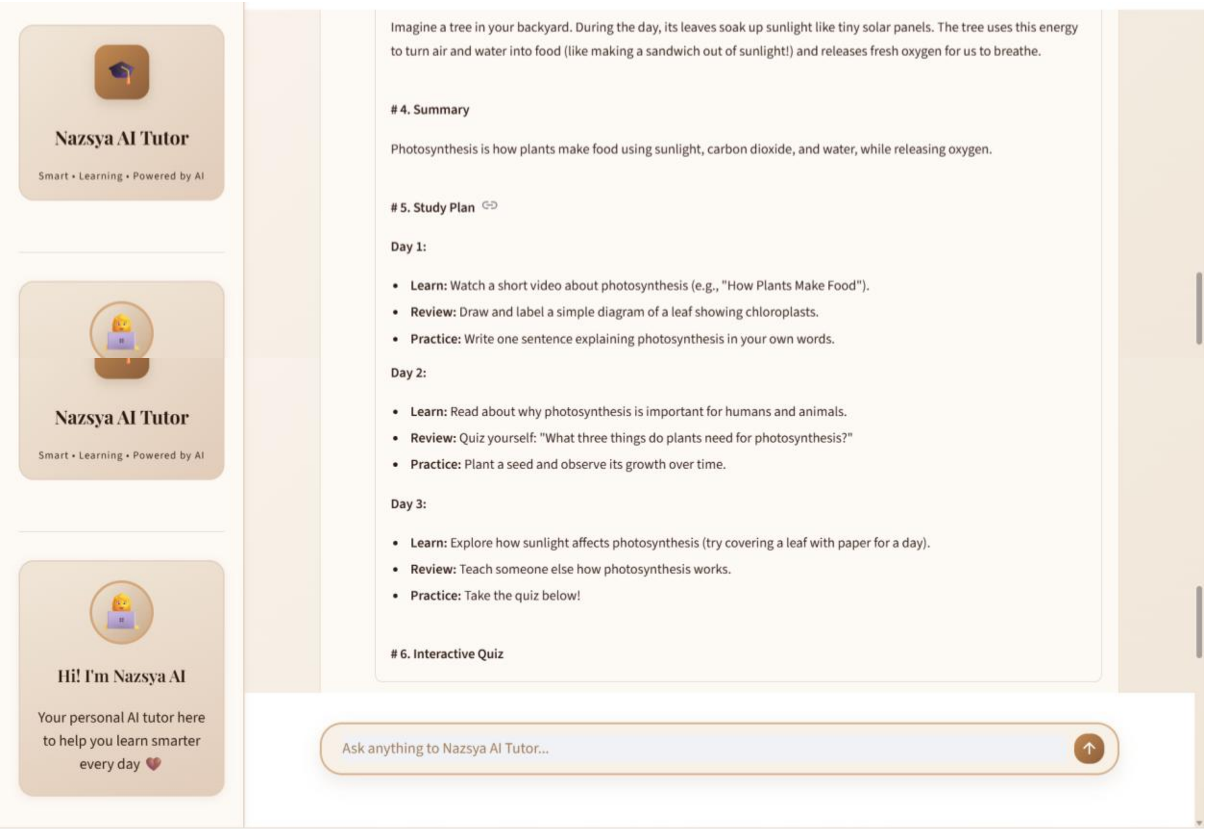



**Figure 1.5** *Final comparison display between Basic Prompt Response and Optimized Prompt Response*

### Summary of prompt comparison output:

The Basic Prompt Response generated a general explanation of photosynthesis. It explained the definition, equation, importance, and main stages of photosynthesis. Although the response was useful, it was still presented as a general explanation and did not provide a complete learning structure.

The Optimized Prompt Response generated a more organized educational response. It included structured sections such as Definition, Detailed Explanation, Real-World Example, Summary, and Study Plan. The explanation was also more learner-friendly and used simple analogies to make the topic easier to understand.

This comparison shows that optimized prompting improved the structure, readability, and educational usefulness of the AI-generated response compared to the basic prompt.

## 1.4 Prompt Comparison Analysis

The prompt comparison experiment demonstrated that different prompt engineering strategies significantly affected the quality of AI-generated educational responses.

The Basic Prompt Response produced a short and general explanation of the topic. It was useful for quick understanding, but it did not include structured sections, learner-level adaptation, study plans, quizzes, or interactive evaluation.

In contrast, the Optimized Prompt Response generated a more complete educational output. It included Definition, Detailed Explanation, Real-World Example, Summary, Study Plan, and

Interactive Quiz. The optimized response was also adapted based on the selected learner level and evaluator role.

This shows that optimized prompting improved response structure, completeness, personalization, and educational usefulness compared to basic prompting.

The following table summarizes the comparison between the Basic Prompt Response and the Optimized Prompt Response in the final application.

| Aspect | Basic Prompt Response | Optimized Prompt Response |
|---|---|---|
| Response Style | Short and general | Structured and educational |
| Output Format | Paragraph-based explanation | Definition, Explanation, Example, Summary, Study Plan, and Quiz |
| Learner Level Adaptation | Not available | Available |
| Study Plan | Not available | Available |
| Quiz | Not available | Available |
| Educational Usefulness | Basic understanding | Complete learning support |
| Prompt Engineering Technique | Basic Prompting | Optimized Prompting with multiple techniques |

Based on the prompt comparison engine testing, the project concluded that optimized prompting produced more structured, complete, and educationally useful responses than basic prompting. The comparison engine helped demonstrate the impact of advanced prompt engineering techniques in improving the quality of AI-generated learning content.

## 2. Prompt Optimization

In this phase, the project optimized the prompt structure to improve educational response quality, clarity, organization, consistency, and personalization.

The optimization process was carried out by adding clearer instructions, structured response sections, learner-level adaptation, few-shot prompting examples, study plan generation, quiz generation, and role-based behavior rules.

In the final OpenRouter-based implementation, the optimized prompt guided the DeepSeek Chat model to generate more complete and educational responses. The optimized response was designed not only to explain a topic, but also to support learning through examples, summaries, study plans, quizzes, and role-based instructional guidance.

### 2.1 Initial Optimized Prompt Implementation

The following code shows the initial optimized prompt design used to improve the structure and clarity of AI-generated educational responses.

```
optimized_prompt = f"""
You are an expert AI educational tutor.
```

```
Your task is to provide:
- Clear explanations
- Beginner-friendly educational content
- Step-by-step explanations
- Concise and well-structured answers
- Real-world examples
- Simple summaries

Explain the following topic for a {level} learner:

Topic:
{question}

Format:
1. Definition
2. Detailed Explanation
3. Real-World Example
4. Summary

Keep the explanation educational, organized, and easy to understand.
"""
```


**Figure 2.1** *Optimized prompt implementation for improving educational response quality*

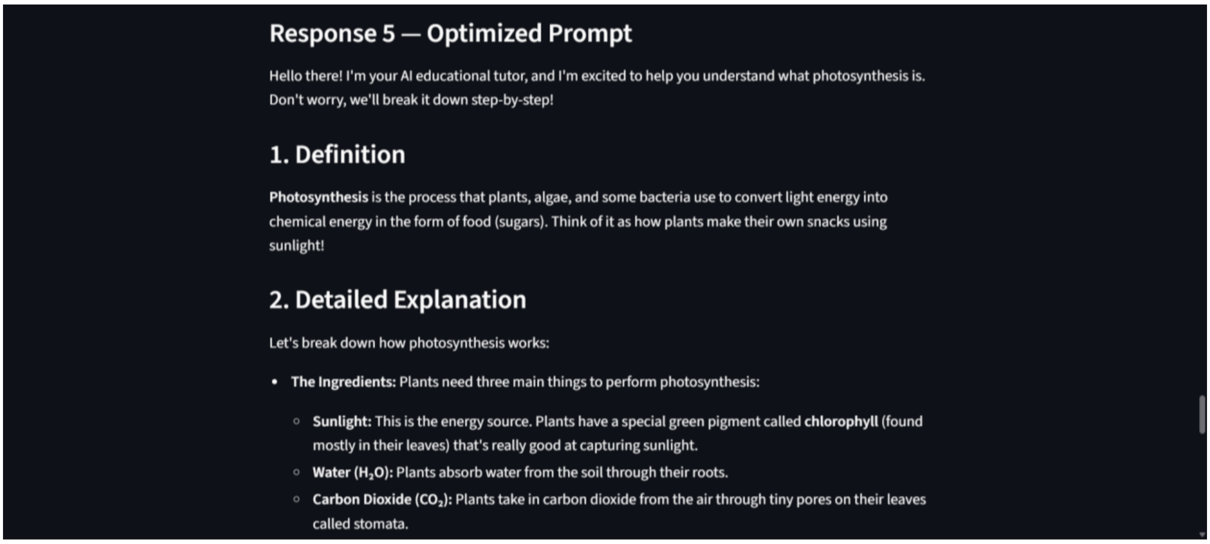


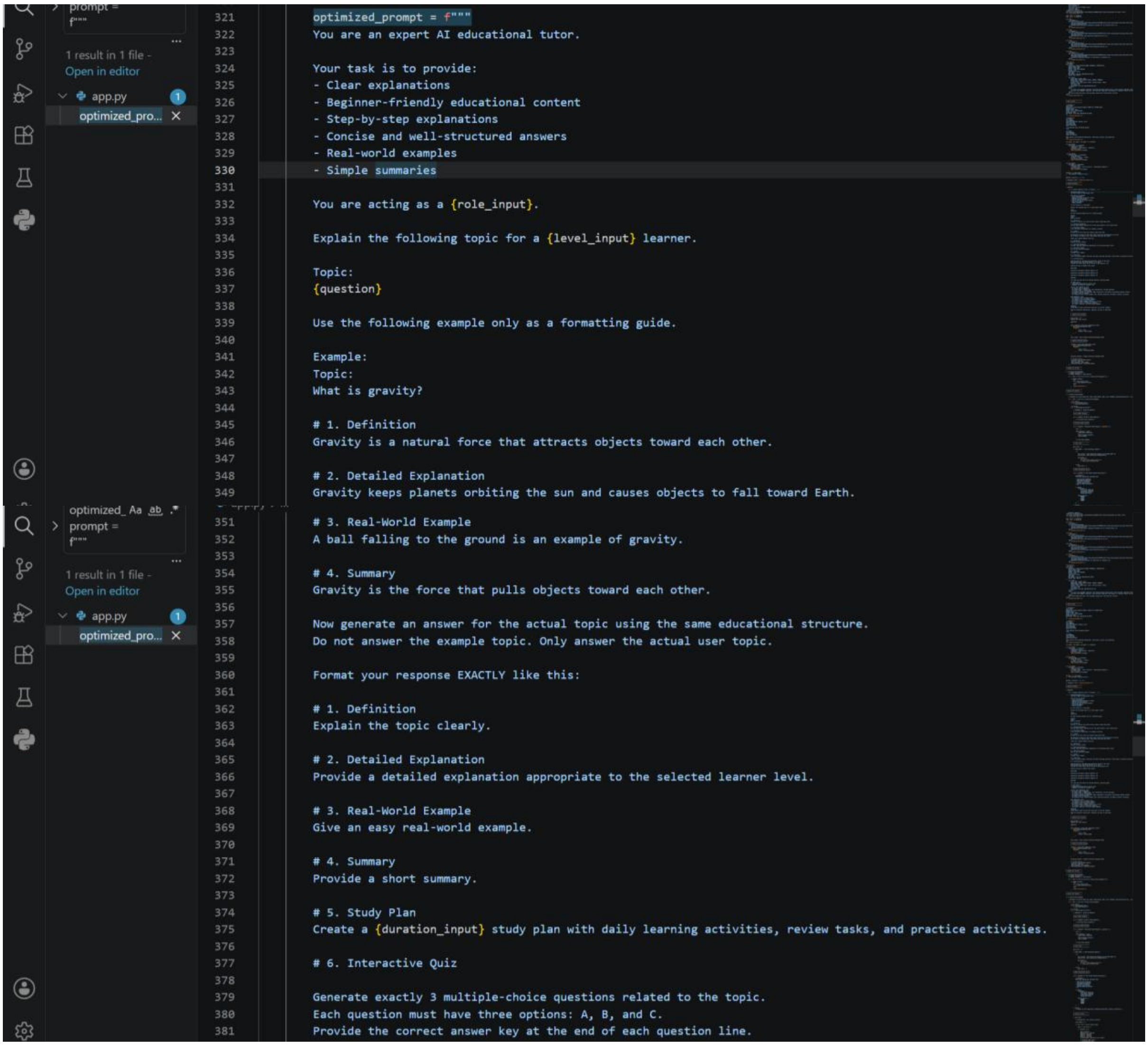

**Figure 2.2** *AI-generated educational response using optimized prompt engineering techniques*

## 2.2 Final Optimized Prompt Implementation in OpenRouter Version

After the initial optimized prompt was tested, the prompt structure was further improved and implemented in the final version of Nazsya AI Tutor using OpenRouter API and the DeepSeek Chat model.

The final optimized prompt combined several prompt engineering techniques, including structured prompting, few-shot prompting, learner-level adaptation, study plan generation, interactive quiz generation, and role-based instruction.

This final prompt allowed the system to generate a more complete educational response. The output included Definition, Detailed Explanation, Real-World Example, Summary, Study Plan, and Interactive Quiz. The response style was also adjusted based on the selected learner level and evaluator role.



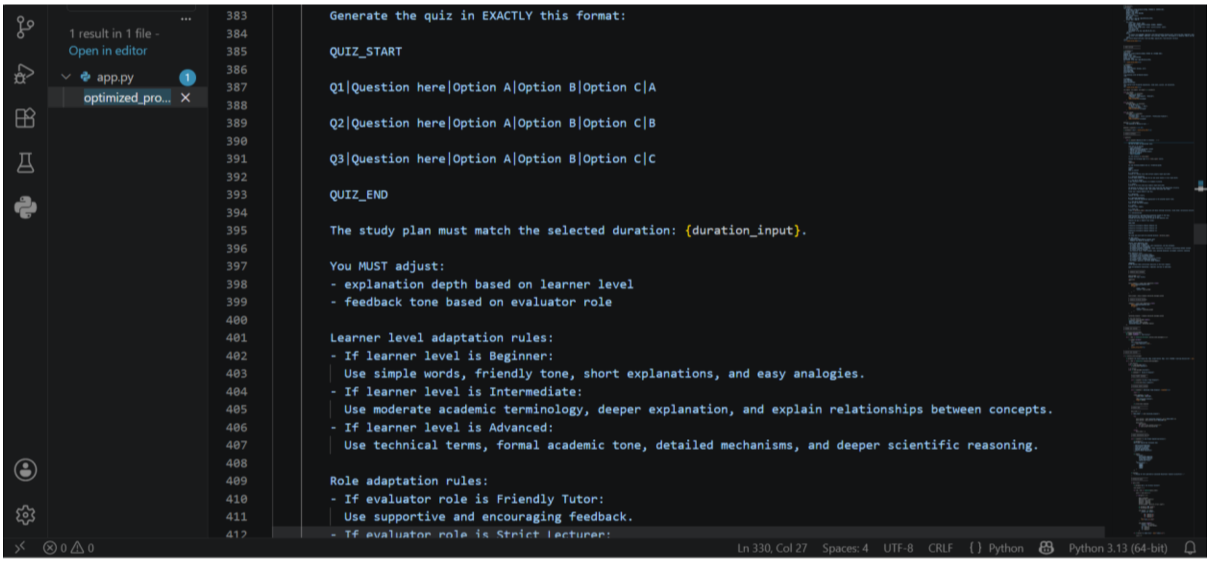



**Figure 2.3** *Final optimized prompt implementation using OpenRouter API*

## 2.3 Prompt Optimization Analysis

The optimized prompt produced more educational, structured, and personalized AI responses compared to the basic prompt.

The final optimized prompt improved response quality by giving the AI clear instructions about output format, learner level, study duration, quiz format, and response style. As a result, the generated response became more organized and useful for learning.

Compared to the basic prompt, the optimized prompt produced a more complete educational output because it included structured explanations, real-world examples, summaries, study plans, and interactive quiz content.

| Aspect | Basic Prompt | Final Optimized Prompt |
|---|---|---|
| Structure | General explanation | Highly structured educational format |
| Educational Clarity | Moderate | High |
| Readability | Good | Very good |
| Example Quality | Basic | Real-world contextual example |
| Learner Adaptation | Not available | Available |
| Study Plan | Not available | Available |
| Quiz Generation | Not available | Available |
| Few-Shot Guidance | Not available | Available |
| Consistency | Moderate | High |

Based on the prompt optimization testing, the project concluded that the final optimized prompt improved the quality of AI-generated educational responses. The optimized prompt produced more structured, complete, consistent, and learner-friendly outputs compared to the basic prompt. This shows that prompt optimization plays an important role in improving the performance of Nazsya AI Tutor as an AI-powered educational assistant.

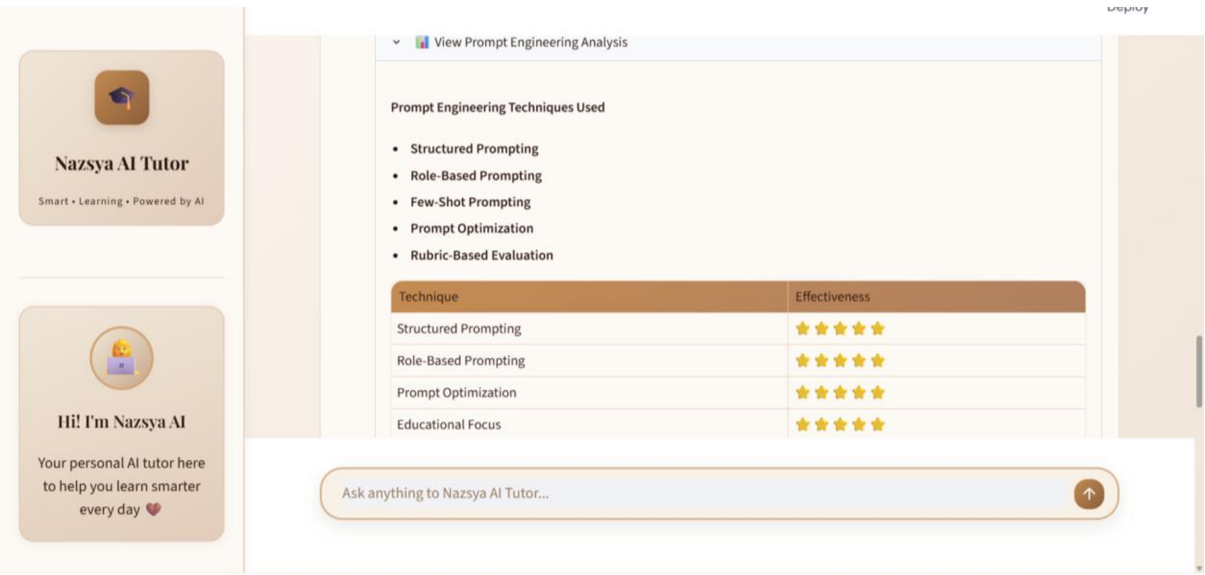



**Figure 2.4** *Prompt engineering analysis of techniques used in the final Nazsya AI Tutor application*

The prompt engineering analysis section shows the main techniques implemented in the final Nazsya AI Tutor application, including Structured Prompting, Role-Based Prompting, Few-Shot Prompting, Prompt Optimization, and Rubric-Based Evaluation. This section helps summarize the prompt engineering strategies used to improve the quality of AI-generated educational responses.

### 3. Performance Evaluation

In this phase, the project evaluated the effectiveness of multiple prompt engineering techniques implemented in Nazsya AI Tutor. The evaluation focused on educational response quality metrics, including clarity, structure, and consistency.

The purpose of this evaluation was to compare how different prompting strategies affected the quality of AI-generated educational responses. The evaluated prompt techniques included Standard Prompting, Basic Prompting, Structured Prompting, Few-Shot Prompting, and Optimized Prompting.

The evaluation results showed that Optimized Prompting generated the highest overall educational response quality because it combined multiple prompt engineering techniques into one complete educational response format.

### 3.1 Performance Evaluation Implementation

The following code shows the performance evaluation implementation used to compare the effectiveness of several prompt engineering techniques.

The evaluation table compares five prompt techniques: Standard Prompt, Basic Prompt, Structured Prompt, Few-Shot Prompt, and Optimized Prompt. Each technique was evaluated based on three criteria: Clarity, Structure, and Consistency.

```
st.subheader("Performance Evaluation Results")
```

```
st.table({
    "Prompt Technique": [
        "Standard Prompt",
        "Basic Prompt",
        "Structured Prompt",
        "Few-Shot Prompt",
        "Optimized Prompt"
    ],

    "Clarity": [
        "Moderate",
        "Moderate",
        "High",
        "High",
        "Very High"
    ],

    "Structure": [
        "Moderate",
        "Low",
        "High",
        "Moderate",
        "Very High"
    ],

    "Consistency": [
        "Moderate",
        "Moderate",
        "High",
        "Very High",
        "Very High"
    ]
})
```

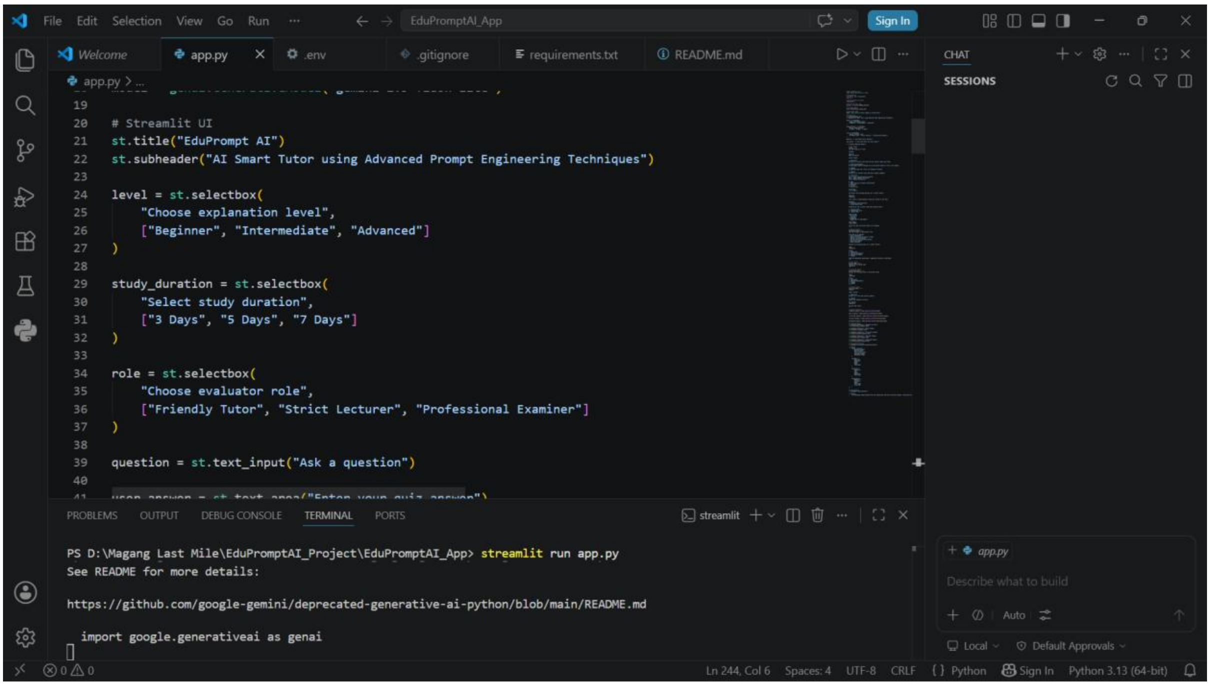



**Figure 3.1** *Performance evaluation implementation for prompt engineering comparison*

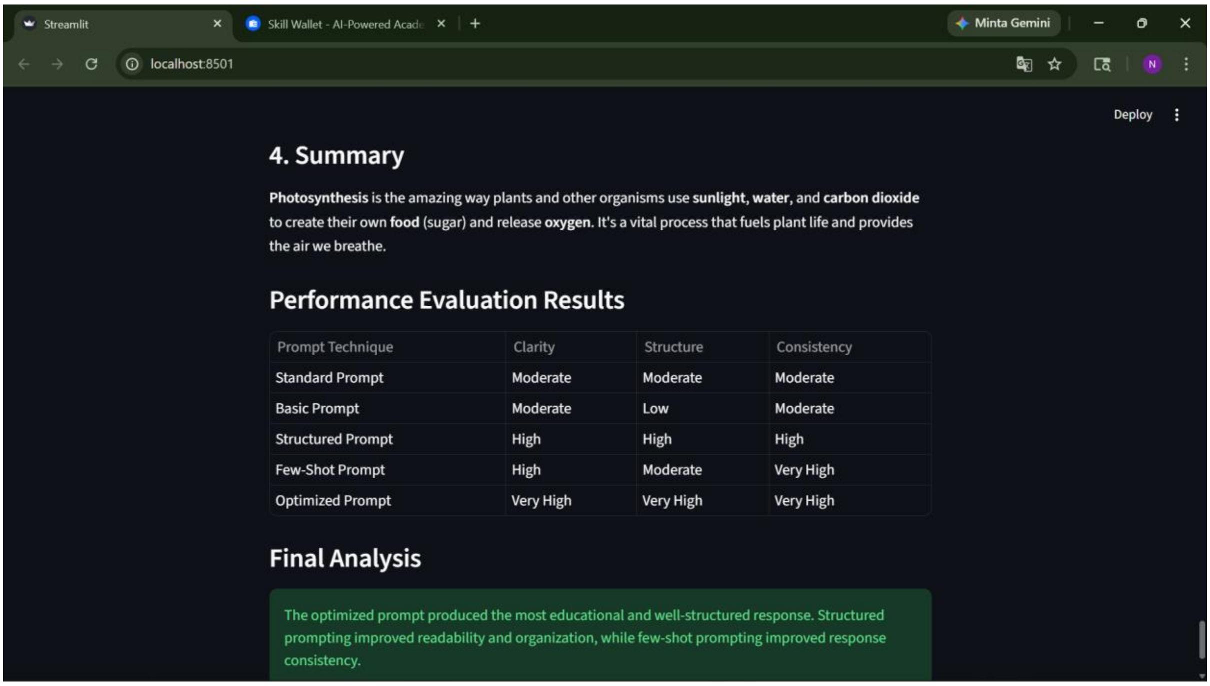



**Figure 3.2** *Performance evaluation results and final analysis of multiple prompt engineering techniques*

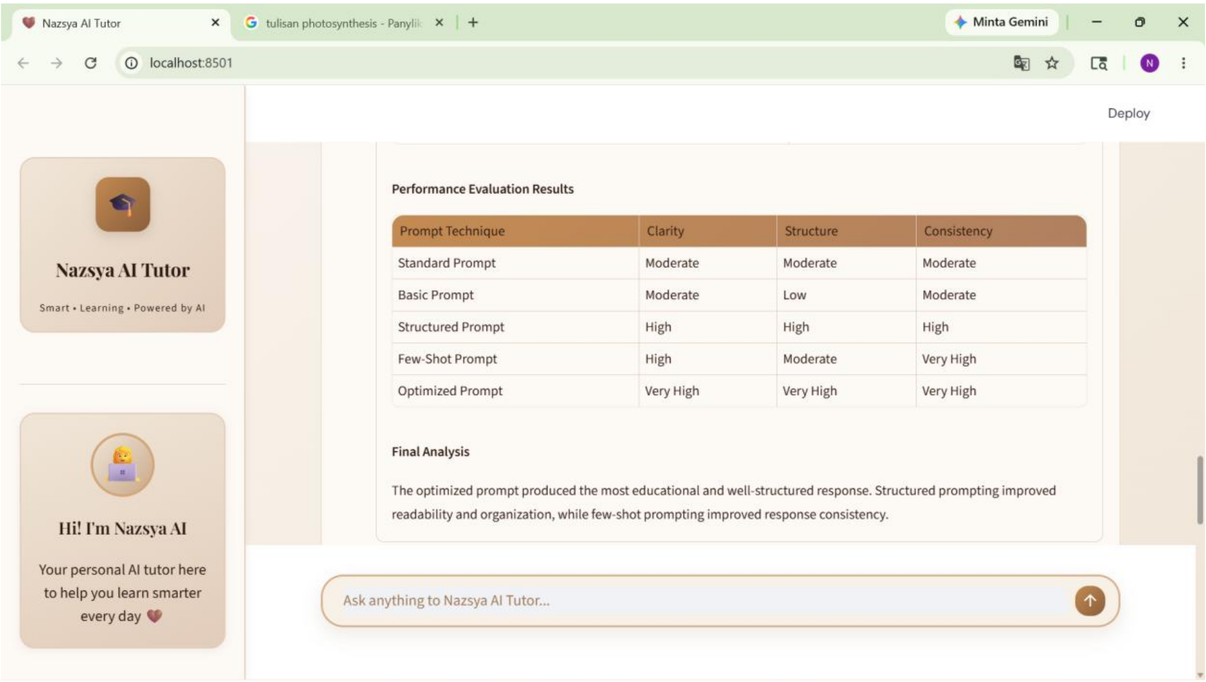



**Figure 3.3** *Performance evaluation results of multiple prompt engineering techniques*

**Summary of performance evaluation output:**

The performance evaluation compared several prompt engineering techniques, including Standard Prompting, Basic Prompting, Structured Prompting, Few-Shot Prompting, and Optimized Prompting.

The Standard Prompt generated a general educational response, while the Basic Prompt produced a longer but less structured explanation. The Structured Prompt improved organization by separating the response into clearer sections. The Few-Shot Prompt produced a more consistent format by following the provided example.

The Optimized Prompt produced the strongest result because it combined structured explanation, learner-friendly language, real-world examples, summary, study plan, and quiz generation. Based on the evaluation table, the Optimized Prompt achieved Very High ratings in clarity, structure, and consistency.

### 3.2 Performance Evaluation Analysis

The performance evaluation results confirmed that prompt engineering significantly influenced AI-generated educational responses.

Standard Prompting produced a general response, while Basic Prompting generated a longer explanation but with weaker structure. Structured Prompting improved response organization and readability. Few-Shot Prompting improved consistency because the AI followed the provided example format.

Optimized Prompting produced the best overall result because it combined several techniques, including structured prompting, few-shot prompting, learner-level adaptation, study plan generation, quiz generation, and role-based evaluation.

Based on the performance evaluation table, the Optimized Prompt achieved Very High ratings in clarity, structure, and consistency. This shows that prompt optimization improved the instructional effectiveness and educational quality of Nazsya AI Tutor.

The following table summarizes the best-performing prompt technique based on each evaluation aspect.

| **Evaluation Aspect** | **Best Performing Prompt** | **Reason** |
|---|---|---|
| Clarity | Optimized Prompt | Provides clear and learner-friendly explanation |
| Structure | Optimized Prompt | Uses organized sections such as Definition, Explanation, Example, Summary, Study Plan, and Quiz |
| Consistency | Optimized Prompt | Follows a stable prompt format and output structure |
| Educational Usefulness | Optimized Prompt | Supports explanation, practice, study planning, and evaluation |
| Personalization | Optimized Prompt | Adapts response based on learner level and evaluator role |

Based on the performance evaluation, the project concluded that optimized prompting produced the highest educational response quality compared to other prompt engineering techniques. The optimized prompt improved clarity, structure, consistency, and overall learning usefulness.

This evaluation demonstrated that advanced prompt engineering techniques play an important role in improving the performance of Nazsya AI Tutor as an AI-powered educational assistant.

Overall, the Prompt Comparison Engine successfully demonstrated the difference between basic prompting and optimized prompting in Nazsya AI Tutor. The comparison results showed that optimized prompting produced clearer, more structured, more consistent, and more educationally useful responses. Therefore, this optional component supports the main objective of the project by proving that advanced prompt engineering techniques can improve AI-generated learning content.